

Cybersecurity Project 1 - Home Lab Setup.

1. Objective: To set up a safe, isolated home lab environment using VirtualBox with Kali Linux (attacker) and Debian (victim) machines, communicating via a host-only network.

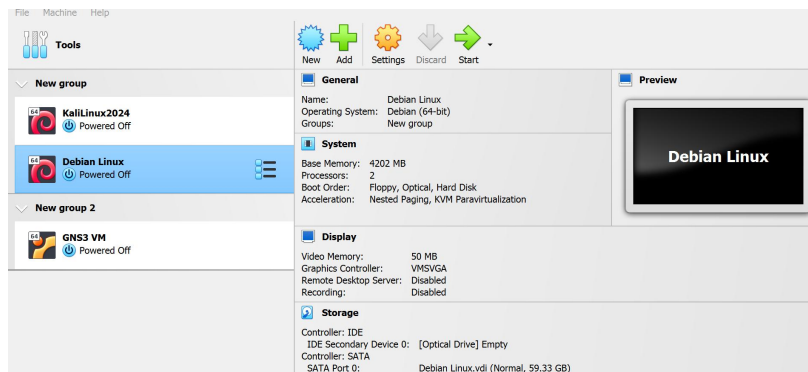
2. Tools Used:

- Oracle VirtualBox
- Kali Linux 2024 ISO
- Debian Linux ISO
- Host-only Adapter (VirtualBox Networking)

3. Steps:

Step 1: Install Kali Linux and Debian VMs

- Download Kali and Debian ISO files.
- Create two VMs in VirtualBox: one for Kali, one for Debian.
- Assign each VM:
 - 2 GB RAM
 - At least 20 GB virtual hard disk
- Mount the ISO and continue with the setup process.



Step 2: Configure Network Adapters

Go to **Settings > Network** for both **Kali Linux** and **Debian** VMs in VirtualBox and configure the following:

Adapter 1 (NAT)

- ✧ **Attached to:** NAT
- ✧ Purpose: Provides internet access to download packages (e.g., Apache, updates).

Adapter 2 (Host-only)

- ✧ **Attached to:** Host-only Adapter
- ✧ **Name:** VirtualBox Host-Only Ethernet Adapter
- ✧ Purpose: Allows private network communication between the Kali and Debian VMs (isolated from internet and host system).

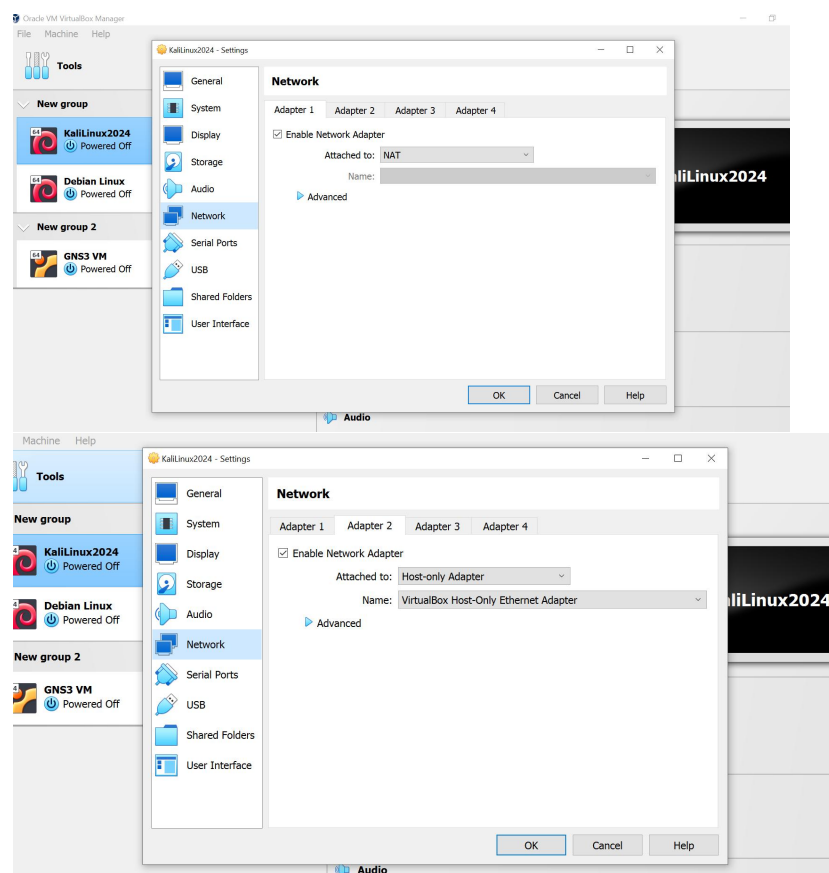


Figure 1 Kali Network configurations

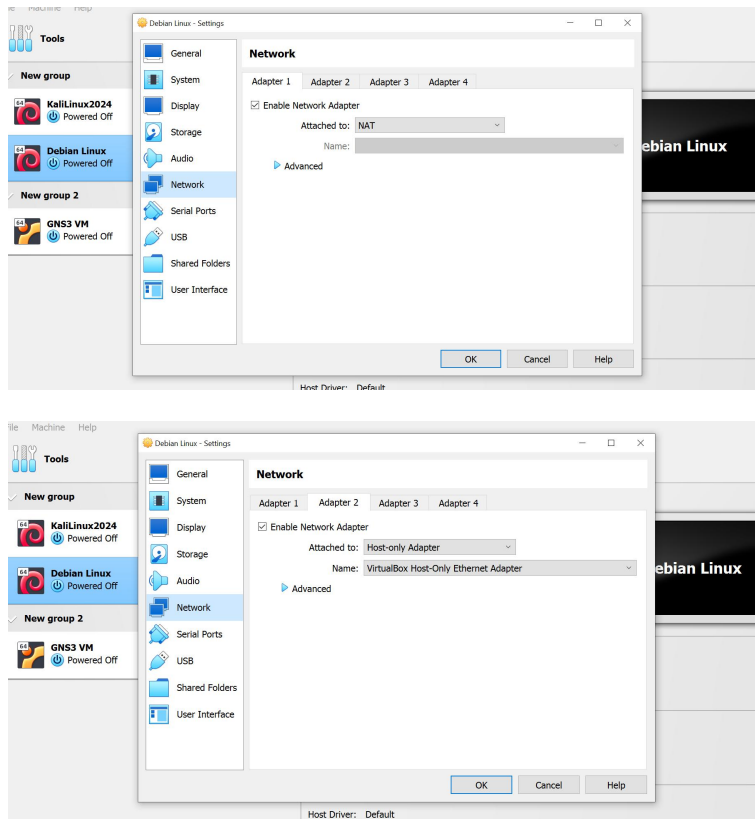


Figure 2 Debian Network configurations

Step 3: Set Static IPs on Both Machines

Debian:

`sudo nano /etc/network/interfaces`

Add:

**`auto enp0s8
iface enp0s8 inet static
address 192.168.56.104
netmask 255.255.255.0`**

Kali:

`sudo nano /etc/network/interfaces`

Add:

**`auto eth1
iface eth1 inet static
address 192.168.56.105
netmask 255.255.255.0`**

Then:

sudo systemctl restart networking

Debian Linux [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
Activities Terminal May 27 20:57

miski@Misky:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1c:0f:93 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.11/24 brd 192.168.10.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe1c:f93/64 scope link
        valid_lft forever preferred_lft forever
3: ens8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:00:c2:75 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic noprefixroute ens8
        valid_lft 371sec preferred_lft 371sec
    inet6 fe80::a00:27ff:fe00:c275/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
miski@Misky:~$
```

```
root@kali: ~
File Actions Edit View Help
root@kali:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1c:0f:93 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.11/24 brd 192.168.10.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe1c:f93/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:60:e5:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 555sec preferred_lft 555sec
    inet6 fe80::a00:27ff:fe60:e59c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@kali:~#
```

Step 4: Test Communication

From Kali:

ping 192.168.56.104

```
root@kali: ~
File Actions Edit View Help
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:60:e5:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 555sec preferred_lft 555sec
    inet6 fe80::a00:27ff:fe60:e59c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@kali:~# ping 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data:
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=0.651 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=0.583 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=0.553 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=0.451 ms
64 bytes from 192.168.56.104: icmp_seq=5 ttl=64 time=0.703 ms
64 bytes from 192.168.56.104: icmp_seq=6 ttl=64 time=0.471 ms
64 bytes from 192.168.56.104: icmp_seq=7 ttl=64 time=0.371 ms
^C
--- 192.168.56.104 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6107ms
rtt min/avg/max/mdev = 0.371/0.540/0.703/0.108 ms
root@kali:~#
```

```
ping 192.168.56.105
```



```
sudo apt update
sudo apt install apache2
```



Step 6: Access Web Page from Kali

In Kali Firefox:

<http://192.168.56.104>

You should see the **Apache2 Debian Default Page**.

